

Introduction to Artificial Intelligence

Bayesian Networks

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Plan

So far:

- Started exploring probabilistic reasoning
- Introduced probability theory

Today:

- Bayesian Networks (BNs)
- Inference algorithms for computing posteriors probabilities

(Boolean) Bayesian Networks

Recall

Boolean Variable: A Boolean variable is a variable that can take on only two values, typically represented as true (True) and false (False).

Probability Notation:

- $P_X(x) = P(X = x)$ (or $P(x)$ if clear from the context)
- $P(X, Y)$: joint probability of X and Y
- $P(X | Y)$: conditional probability of X given Y
- $P(X) = \sum_y P(X, Y = y)$ (marginalization)

Conditional Independence

Definition: Independence

X and Y are **independent** if

$$\forall x, y : P(x, y) = P(x) P(y)$$

Definition: Conditional Independence

X and Y are **conditionally independent** given Z if

$$\forall x, y, z : P(x, y \mid z) = P(x \mid z) P(y \mid z)$$

(Conditional) independence is a property of a probability distribution.

Building a Probabilistic Model

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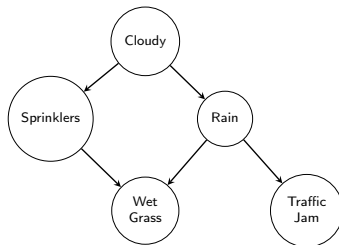
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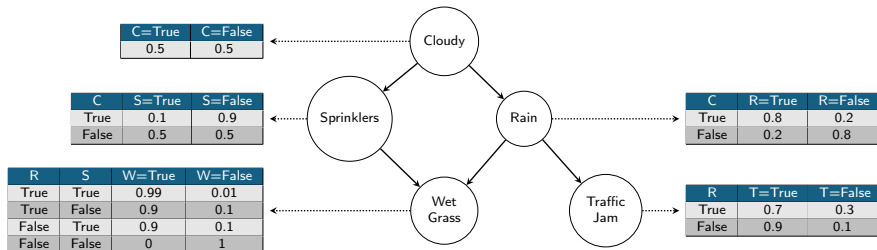


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Bayesian Network – Definition

Definition: Bayesian Network

A **Bayesian Network** is a directed acyclic graph (DAG) in which:

- 1 Nodes represent random variables
- 2 Directed arcs encode “direct influence” (conditional independence)
- 3 Each node X has a **Conditional Probability Table (CPT)**
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Idea:

BN = Topology (graph) + Local Conditional Probabilities

Probabilities in Bayesian Networks

A BN implicitly encodes the full joint distribution as a product of local conditional distributions:

$$P(x_1, x_2, \dots, x_n) = \prod_{i=1}^n P(x_i \mid \text{parents}(x_i))$$

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Example: to know the probability of being cloudy, raining, having sprinklers on, wet grass, and no traffic jam, we can compute:

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Remark:

Conditional independence assumptions allow us to **reconstruct** any entry of the full joint distribution.

Why Reasoning with Bayesian Networks?

Modelling with BNs:

- Build a compact representation of a joint distribution
- Encode conditional independence assumptions explicitly in the graph
- More efficient computations:
 - Full joint over N Boolean variables: 2^N entries
 - BN with at most k parents per node: $O(N \cdot 2^{k+1})$ parameters

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Remark:

BNs preserve the ability to compute joint and conditional probabilities, but with dramatically lower space requirements when $k \ll N$.

Three Canonical Dependency Patterns

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2) **Common cause:** $Y \rightarrow X$ and $Y \rightarrow Z$

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3) **Common effect (collider):** $X \rightarrow Y \leftarrow Z$

Without evidence on Y (or descendants), X and Z are independent; observing Y *activates* dependence.

D-Separation (Graphical Independence Test)

Definition: D-separation

Given evidence set E , two node sets X and Y are d-separated if **every** undirected path between X and Y is blocked.

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EXAMPLE

Blocked Paths: Rules

A path is blocked by evidence E if one of the following holds:

- 1 **Chain** $X \rightarrow Z \rightarrow Y$ or $X \leftarrow Z \leftarrow Y$ and $Z \in E$
- 2 **Fork** $X \leftarrow Z \rightarrow Y$ and $Z \in E$
- 3 **Collider** $X \rightarrow Z \leftarrow Y$ and neither Z nor descendants of Z are in E

Intuition: observed chains/forks block flow; unobserved colliders block flow.

Inference in Bayesian Networks

Inference by Enumeration

Goal:

$$P(Q \mid e) = P(Q, e)$$

Method:

- Expand required terms into sums over hidden variables
- Compute each term from BN factorization
- Normalize at the end

Correct but often expensive due to repeated computation.

Variable Elimination (VE): Idea

Why faster than plain enumeration?

- Enumeration implicitly rebuilds large joint expressions many times
- VE interleaves multiplication and summation to reuse intermediate factors

Idea:

- Start with local CPT factors (instantiate evidence)
- While hidden variables remain:
 - choose a hidden variable H
 - join all factors containing H
 - sum out H

Summary

- BNs provide a compact way to represent complex joint distributions using local conditional probabilities
- The graph structure encodes conditional independence assumptions
- BNs support **predictive**, **diagnostic**, and **intercausal** reasoning
- D-separation tells us when conditional independence holds
- Enumeration is conceptually simple but computationally heavy
- Variable elimination computes exact posteriors more efficiently

Book references: Russell & Norvig, Chapters 12.6, 13.1, 13.2.1, 13.3.(1-3)